Cache-Resident Cross-Section Reconstruction via Singular Value Decomposition for Monte Carlo Neutron Transport

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Abstract

We present a method for on-the-fly reconstruction of neutron cross-sections using truncated Singular Value Decomposition (SVD) that eliminates the memory-wall bottleneck in continuous-energy Monte Carlo transport. The cross-section matrix $\mathbf{A} \in \mathbb{R}^{N_E \times N_T}$, indexed by energy and temperature, is decomposed in log-space as $\log_{10} \mathbf{A} \approx \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^\top$, where the pre-multiplied basis $\mathbf{B} = \mathbf{U}_k \mathbf{\Sigma}_k$ fits entirely within the CPU L3 cache. Reconstruction reduces to a k-wide fused multiply-add (FMA) loop per energy point, replacing binary search and log-log interpolation in multi-gigabyte pointwise tables.

Validation against ENDF/B-VII.1 and JEFF-3.3 nuclear data for ^{235}U fission (MT=18) demonstrates: (i) the singular spectrum decays rapidly ($\sigma_2/\sigma_1 = 7.7 \times 10^{-2}$), confirming low effective rank; (ii) a pure-Rust implementation achieves 8–13× speedup over traditional table lookup at 3–5 ns per energy point; (iii) Godiva (HEU-MET-FAST-001) criticality benchmarks with OpenMC show $\Delta k_{\text{eff}} < 10$ pcm at all tested ranks ($k = 2$ through $k = 5$), indistinguishable from the unmodified library within Monte Carlo statistics. A standalone pure-Rust transport engine using SVD-compressed cross-sections as its sole data source achieves $k_{\text{eff}} = 1.00012 \pm 0.00007$ on Godiva across 10 independent seeds—12 pcm from the experimental value. A head-to-head "honesty test" against an identical engine using OpenMC-style pointwise table lookup shows SVD at 700 ± 8 ns/particle vs table at 832 ± 11 ns/particle during quiet system periods (1.19× speedup) with a 107 pcm fidelity cost at rank 5. A complete event-based GPU transport solver using SVD cross-sections achieves 2.6× speedup over CPU table lookup and 2.1× over CPU SVD on a PWR pin cell benchmark (8 nuclides, S(α,β) thermal scattering) on a laptop RTX A1000 GPU, with physics parity confirmed by matching k_∞ to within Monte Carlo statistics. SVD inherently solves the GPU memory coalescing problem that limits conventional pointwise table lookup: the rank-k dot product reads contiguous memory, eliminating warp divergence and enabling streaming access patterns that GPUs are optimised for. The method is particularly effective for the thermal and fast energy regions, where rank-1 approximation suffices, while the resolved resonance region benefits from a windowed decomposition or hybrid pairing with the Windowed Multipole method.

Keywords: Monte Carlo, cross-section compression, singular value decomposition, cache optimization, nuclear data, GPU transport, CUDA, Rust

1 Introduction

Continuous-energy Monte Carlo neutron transport codes such as OpenMC [1], MCNP, and Serpent rely on pointwise cross-section tables spanning 10^4–10^5 energy points per nuclide and temperature. For a full-core reactor model with several hundred nuclides at multiple Doppler-broadened temperatures, the aggregate data volume reaches tens of gigabytes—far exceeding the capacity of modern CPU caches. Every cross-section lookup thus incurs a cache miss to main memory (100–200 ns latency), creating a *memory wall* that dominates the runtime of particle tracking.

To quantify the scale of the problem: the ENDF/B-VII.1 library contains 444 nuclide files totalling 5.8 GB on disk (the newer ENDF/B-VIII.0 is 13.7 GB due to denser energy grids). Table 1 shows the in-memory footprint for representative nuclides when loaded by OpenMC. A single actinide such as ^{238}U requires 85.5 MB of pointwise data (51 reactions $\times$ 6 temperatures, 5.6 million data points). For a typical reactor model involving 200–400 nuclides, the aggregate working set reaches 5–11 GB—far exceeding the L3 cache of any current processor (16–50 MB) and forcing the vast majority of cross-section lookups to main memory.

Table 1: In-memory pointwise data size per nuclide (ENDF/B-VII.1, 6 temperatures). Data points include energy grid + cross-section value (16 bytes per point).

Nuclide	Role	Reactions	Data points	RAM
^{235}U	Fissile actinide	52 $\times$ 6T	2.6 M	39.7 MB
^{238}U	Fertile actinide	51 $\times$ 6T	5.6 M	85.5 MB
^{239}Pu	Fissile actinide	52 $\times$ 6T	2.5 M	38.1 MB
^{16}O	Moderator/fuel	72 $\times$ 6T	0.3 M	4.3 MB
^{1}H	Moderator	6 $\times$ 6T	0.02 M	0.3 MB
^{90}Zr	Structural	52 $\times$ 6T	0.15 M	2.2 MB
6 nuclides (PWR pin cell)				170 MB
Projected: 50 nuclides				1.4 GB
Projected: 200 nuclides				5.5 GB
Projected: 400 nuclides				11.1 GB

Several strategies have been proposed to mitigate this bottleneck. Machine-learning-based adaptive resampling reduces pointwise grid density while preserving resonance features, achieving $\Delta k_{\text{eff}} < 100$ pcm with 10–50% of the original data [2, 3]. The Windowed Multipole (WMP) method represents cross-sections analytically using multipole parameters, reducing memory by one to two orders of magnitude for the resolved resonance region [4, 5]. Hybrid neural-network approaches combine WMP with learned Doppler broadening kernels, achieving 65% memory reduction [6].

In this work we propose a complementary approach inspired by low-rank compression techniques from the AI/ML community—specifically the KV-cache compression methods used in large language models [7, 8, 9, 10]. We observe that the cross-section matrix, when viewed in log-space, exhibits rapid singular-value decay, and exploit this structure via truncated SVD to build a *cache-resident* reconstruction kernel.

The key contributions are:

1. A pure-Rust implementation that reads OpenMC HDF5 nuclear data, performs

SVD decomposition, and reconstructs cross-sections entirely without C/Fortran dependencies.

2. Empirical characterisation of the singular spectrum for ^{235}U, ^{238}U, and ^{239}Pu across two independent nuclear data libraries (ENDF/B-VII.1 and JEFF-3.3).

3. Demonstration that the SVD basis vectors fit in L3 cache, enabling reconstruction via streaming FMA at 3–5 ns per energy point—an order of magnitude faster than table lookup.

4. Godiva criticality benchmark validation showing $\Delta k_{\text{eff}} < 10$ pcm, within Monte Carlo statistical noise.

5. A windowed SVD analysis showing that the thermal ($E < 1$ eV) and fast ($E > 25$ keV) regions are effectively rank-1, while the resolved resonance region requires $k = 5$–6 for sub-percent accuracy.

6. A standalone pure-Rust Monte Carlo transport engine using SVD cross-sections with full physics (anisotropic scattering, tabulated fission spectra, URR probability tables, free-gas thermal model) achieving $k_{\text{eff}} = 1.00012 \pm 0.00007$ on Godiva (10 independent seeds, 12 pcm from experiment).

7. A statistically rigorous head-to-head comparison ("honesty test") against pointwise table lookup in the same engine, reporting 1.19× speedup in ns/particle during quiet system periods (700 ± 8 vs 832 ± 11 ns/particle) with 107 pcm fidelity cost, benchmarked over 10 seeds each.

8. Resonance integral validation confirming $< 0.2\%$ error on $\int \sigma(E)\, dE/E$ across all standard energy groups.

9. GPU reconstruction kernel achieving 4.2× speedup over CPU with zero error (machine epsilon).

10. A complete event-based GPU transport solver (Tramm architecture) with physics parity, achieving 2.6× speedup over CPU on a PWR pin cell with 8 nuclides and S(α,β) thermal scattering.

2 Method

2.1 Problem formulation

Let $\sigma(E, T)$ denote the microscopic cross-section (barns) for a given nuclide and reaction, evaluated on a discrete energy grid $\{E_i\}_{i=1}^{N_E}$ at temperatures $\{T_j\}_{j=1}^{N_T}$. We form the matrix

$$\mathbf{A} \in \mathbb{R}^{N_E \times N_T}, \quad A_{ij} = \sigma(E_i, T_j), \tag{1}$$

and work in log-space to normalise the six-order-of-magnitude dynamic range:

$$\tilde{\mathbf{A}} = \log_{10} \mathbf{A}. \tag{2}$$

2.2 SVD decomposition

The thin SVD of $\tilde{\mathbf{A}}$ yields

$$\tilde{\mathbf{A}} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top, \tag{3}$$

where $\mathbf{U} \in \mathbb{R}^{N_E \times r}$, $\boldsymbol{\Sigma} = \mathrm{diag}(\sigma_1, \ldots, \sigma_r)$, $\mathbf{V} \in \mathbb{R}^{N_T \times r}$, and $r = \min(N_E, N_T)$.

The rank-k truncation retains the first k singular triplets:

$$\tilde{\mathbf{A}}_k = \mathbf{U}_k\boldsymbol{\Sigma}_k\mathbf{V}_k^\top, \quad k \ll r. \tag{4}$$

2.3 Cache-resident kernel

For the reconstruction hot path, we pre-multiply the singular values into the basis:

$$\mathbf{B} = \mathbf{U}_k\boldsymbol{\Sigma}_k \in \mathbb{R}^{N_E \times k}, \quad B_{ij} = U_{ij} \cdot \sigma_j. \tag{5}$$

At runtime, for a given temperature T corresponding to column index t:

1. Compute the coefficient vector $\mathbf{c}(t) = \mathbf{V}_k^\top \mathbf{e}_t \in \mathbb{R}^k$ (one cache line, computed once per temperature change).

2. For each energy point i, reconstruct:

$$\log_{10}\sigma(E_i, T) = \sum_{j=1}^{k} B_{ij} \cdot c_j. \tag{6}$$

Equation (6) is a dot product of length k, compiled to k fused multiply-add (FMA) instructions on x86. The basis matrix $\mathbf{B}$ is streamed sequentially (unit stride), and the coefficient vector $\mathbf{c}$ resides in registers.

Energy index lookup. The energy index i must be determined before reconstruction. The SVD kernel speedup applies to the *reconstruction step* (Eq. 6), not to the energy search. In the full transport engine, we apply two optimisations: (i) a single search shared across all reaction channels within a nuclide (which use the same unionised energy grid via reference counting), eliminating 5 of 6 redundant searches per collision; and (ii) a hash-based $O(1)$ index lookup [12] using 8192 log-uniform bins, replacing the $O(\log N_E)$ binary search ($\sim$17 comparisons for $N_E \approx 10^5$).

Memory analysis. For ^{235}U fission with $N_E = 83{,}114$ and $k = 5$:

- $\mathbf{B}$: $83{,}114 \times 5 \times 8$ bytes $= 3.2$ MB (fits in L3 cache).
- $\mathbf{c}$: 5×8 bytes $= 40$ bytes (fits in a single cache line).
- Original pointwise table: $83{,}114 \times 6 \times 2 \times 8$ bytes $= 7.8$ MB per reaction (energy + xs arrays).

2.4 Windowed decomposition

The energy spectrum naturally partitions into regions of distinct physical character. We apply independent SVD to each window:

- Thermal ($E < 1$ eV): smooth $1/v$ behaviour, effective rank 1.
- Resolved resonances (1 eV–25 keV): Breit-Wigner peaks, requires $k = 5$–6.
- Fast ($E > 25$ keV): smooth, effective rank 1.

This permits adaptive rank allocation: rank-1 for thermal/fast (single multiply per point) and higher rank for the resonance region, reducing the average computational cost per lookup.

3 Implementation

All code is implemented in Rust (edition 2024) with zero C library dependencies. Key components:

- **HDF5 reader**: The `hdf5-pure` crate (v0.1.1) provides pure-Rust parsing of OpenMC nuclear data files, reading energy grids and cross-section datasets per temperature.
- **SVD computation**: The `faer` crate (v0.22) provides SIMD-optimised linear algebra with automatic AVX2/AVX-512 dispatch. SVD of an $83{,}114 \times 8$ matrix completes in 20 ms.
- **Reconstruction kernel**: Hand-written FMA loop (Sec. 2.3) using `f64::mul_add` for guaranteed fused multiply-add codegen.
- **Validation**: Bit-exact agreement with OpenMC's Python API (`openmc.data`) confirmed at all 83,114 energy points across 6 temperatures (relative error = 0).

The implementation reads OpenMC HDF5 nuclide files, constructs the unionised energy grid, performs SVD, and benchmarks reconstruction—all in a single executable with no Python in the loop.

Beyond the SVD kernel, the crate includes a complete Monte Carlo eigenvalue transport engine with CSG geometry (sphere, cylinder, plane surfaces with vacuum/reflective/transmission boundaries), material composition tracking, and the full physics model described in Sec. 4.12. Transport is parallelised over source particles within each batch using the `rayon` crate. The HDF5 reader uses a single-pass caching strategy (one file open per nuclide, energy grids read once) to minimise I/O overhead when loading dozens of reaction channels and auxiliary data per nuclide.

4 Results

4.1 Singular spectrum

Table 2 shows the singular values for ^{235}U fission (MT=18) from the JEFF-3.3 combined library (8 temperatures: 250–2500 K).

The ratio $\sigma_2/\sigma_1 = 7.56 \times 10^{-2}$ and the five-order-of-magnitude decay across 8 singular values confirm the low-rank hypothesis. Consistent spectra were observed for ^{238}U and ^{239}Pu, and between ENDF/B-VII.1 and JEFF-3.3 evaluations.

Table 2: Singular spectrum of the ^{235}U fission cross-section matrix in log-space (JEFF-3.3, 8 temperatures).

k	σ_k	σ_k/σ_1	Cumulative energy (%)
1	8.732×10^2	1.000	99.375
2	6.602×10^1	7.56×10^{-2}	99.964
3	1.441×10^1	1.65×10^{-2}	99.998
4	3.383	3.87×10^{-3}	99.99993
5	8.600×10^{-1}	9.85×10^{-4}	99.999996
6	2.953×10^{-1}	3.38×10^{-4}	99.9999999

4.2 Reconstruction accuracy

Table 3 reports the maximum relative error of SVD reconstruction versus the original cross-section data, broken down by energy region.

Table 3: Maximum relative error of rank-k SVD reconstruction for ^{235}U fission (ENDF/B-VII.1, validated against OpenMC).

k	Thermal (< 1 eV)	Resonance (1 eV–25 keV)	Fast (> 25 keV)	Overall
2	3.8×10^{-2}	3.4×10^{-1}	3.0×10^{-3}	3.4×10^{-1}
3	3.2×10^{-3}	1.1×10^{-1}	2.2×10^{-4}	1.1×10^{-1}
4	9.7×10^{-4}	2.8×10^{-2}	7.5×10^{-5}	2.8×10^{-2}
5	1.5×10^{-4}	9.9×10^{-3}	4.5×10^{-6}	9.9×10^{-3}
6	$< 10^{-12}$	$< 10^{-12}$	$< 10^{-12}$	$< 10^{-12}$

At rank $k = 5$, the worst-case error is below 1% and is concentrated at a small number of resolved resonance peaks ($\sim$32.8 eV, 31.4 eV). The P99 error in the resonance region is 3.2×10^{-3}, indicating that 99% of resonance points are reconstructed to better than 0.3%.

4.3 Multi-reaction analysis

Table 4 extends the SVD analysis to all three principal reaction channels of ^{235}U.

Table 4: Singular spectrum comparison across reaction types for ^{235}U (ENDF/B-VII.1, 6 temperatures).

MT	Reaction	σ_2/σ_1	k for 99.99%	Max err at $k = 5$
18	(n,f) Fission	7.70×10^{-2}	3	9.9×10^{-3}
2	(n,n) Elastic	1.61×10^{-2}	2	1.3×10^{-3}
102	(n,γ) Capture	1.21×10^{-1}	3	1.8×10^{-2}

Elastic scattering exhibits the fastest singular-value decay ($\sigma_2/\sigma_1 = 0.016$, effective rank 2), reflecting its smooth energy dependence. Radiative capture is the most challenging ($\sigma_2/\sigma_1 = 0.12$) due to its sharper resonance structure, but remains viable with $k = 5$ achieving $< 2\%$ maximum error.

A comprehensive sweep of all 52 reaction channels in ^{235}U reveals that **47 out of 52 reactions are effectively rank-1** ($\sigma_2/\sigma_1 < 10^{-15}$), including all inelastic excitation levels (MT=51–91), (n,2n), (n,3n), and (n,4n). Only 4 reactions fall in Scenario B ($0.01 < \sigma_2/\sigma_1 < 0.1$): elastic, non-elastic total, fission, and damage energy. The sole Scenario C reaction is radiative capture ($\sigma_2/\sigma_1 = 0.12$).

4.4 Hybrid SVD + Windowed Multipole

The resolved resonance region (1 eV–25 keV) contains 99% of the energy grid points but can be handled by the Windowed Multipole (WMP) method using an analytical representation of ∼2 KB per nuclide-reaction [4]. This leaves only the thermal and fast regions (∼800 points, 1% of the grid) for SVD.

Table 5 compares the approaches.

Table 5: Memory comparison: full table vs SVD-only vs hybrid SVD+WMP for ^{235}U fission.

Method	Memory	Max error	Resonance
Full pointwise table	7792 KB	exact	exact
SVD-only $k = 4$	2598 KB	2.8×10^{-2}	2.8×10^{-2}
SVD-only $k = 5$	3247 KB	9.9×10^{-3}	9.9×10^{-3}
Hybrid SVD(k=2)+WMP	15 KB	3.8×10^{-2}	exact (WMP)
Hybrid SVD(k=3)+WMP	21 KB	3.2×10^{-3}	exact (WMP)

The hybrid approach achieves a **530× memory reduction** (15 KB vs 7792 KB) by delegating the resonance region to WMP and handling only the smooth thermal and fast regions with a rank-2 SVD. At rank-3, the thermal/fast error drops to 0.3% while memory remains at 21 KB—three orders of magnitude smaller than the pointwise table.

4.5 Cross-isotope basis sharing

A natural question is whether different nuclides share singular-vector subspaces, which would allow further compression via a common basis (analogous to cross-layer merging in LLM KV-cache compression [9]).

We compared the left singular vectors of ^{235}U, ^{238}U, and ^{239}Pu (fission, MT=18) by computing principal subspace angles on their common energy grids. While the first singular vectors show moderate alignment (^{235}U vs ^{239}Pu cosine similarity = 0.97), the subspace angle jumps to 85° at $k = 2$, indicating that the higher-order resonance shapes are fundamentally different between isotopes.

Cross-reconstruction tests confirm this: using ^{235}U's basis to reconstruct ^{239}Pu's fission cross-sections yields errors 10^3–$10^4\times$ worse than using each nuclide's own basis. **Cross-isotope basis sharing is not viable** for the resolved resonance region—each nuclide requires its own SVD decomposition. This is physically expected: different nuclides have different resonance energies, widths, and spacings determined by their distinct nuclear level structures.

4.6 Windowed SVD

Table 6 summarises the effective rank per energy window.

Table 6: Per-window effective rank (energy for 99.99% cumulative energy) and σ_2/σ_1 ratio for ^{235}U fission.

Window	Energy range	N_E	σ_2/σ_1
Thermal	< 1 eV	461	3.9×10^{-4}
Low reson.	1–100 eV	11,712	6.1×10^{-2}
Mid reson.	100 eV–2.5 keV	69,401	8.3×10^{-2}
High reson.	2.5–25 keV	30	$< 10^{-15}$
Fast	> 25 keV	527	$< 10^{-15}$

The thermal, high-resonance, and fast windows are numerically rank-1. The resonance windows ($\sigma_2/\sigma_1 \approx 0.06$–$0.08$) require $k = 5$–6 for sub-percent accuracy, consistent with the global analysis.

4.7 Reconstruction speed

Table 7 compares the SVD reconstruction kernel against traditional binary-search table lookup, measured on an Intel Xeon (single-threaded, release-optimised Rust).

Table 7: Reconstruction throughput comparison for ^{235}U fission ($N_E = 83{,}114$). Time is per full-spectrum reconstruction.

k	SVD FMA (μs)	Table lookup (μs)	Speedup	Kernel memory (KB)
2	249	3349	13.4×	1948
3	278	3349	12.1×	2598
4	301	3349	11.1×	3247
5	384	3349	8.7×	3896
6	428	3349	7.8×	4546

The speedup arises from two factors: (i) the SVD kernel streams sequentially through the basis matrix (unit stride, cache-friendly), while table lookup requires binary search with logarithmic random access; (ii) reconstruction is pure arithmetic (FMA), while table lookup requires transcendental function evaluation (`ln`, `powf`) for log-log interpolation.

4.8 Criticality benchmark

The Godiva benchmark (ICSBEP HEU-MET-FAST-001) was run using OpenMC 0.15.3 with 10^5 particles per batch and 450 active batches (45 million histories, $\sigma_{\mathrm{MC}} \approx 9$ pcm). Results are shown in Table 8.

4.9 GPU reconstruction

Table 9 reports GPU benchmarks on an NVIDIA RTX A1000 (Ampere, 16 SMs, 1.5 MB L2 cache) for 10^6 simultaneous particle lookups.

The SVD kernel achieves 2.6–2.8× speedup over binary-search table lookup on GPU. Notably, reconstruction time is nearly independent of rank ($k = 2$ to $k = 6$: 1.8–1.9 ms),

Table 8: Godiva k_{eff} with original and SVD-modified ^{235}U fission cross-sections (ENDF/B-VII.1).

Method	k_{eff}	$\pm\sigma$	Δk (pcm)	$> 3\sigma$?
Baseline	0.99857	0.00009	—	—
SVD $k = 2$	0.99848	0.00010	8.9	No
SVD $k = 3$	0.99849	0.00010	7.5	No
SVD $k = 4$	0.99864	0.00009	6.9	No
SVD $k = 5$	0.99842	0.00010	15.0	No

Table 9: GPU reconstruction throughput (10^6 particles, RTX A1000).

Method	Time (ms)	ns/particle	vs. table
Table (binary search)	4.97	5.0	1.0×
SVD $k = 2$	1.80	1.8	2.8×
SVD $k = 4$	1.83	1.8	2.7×
SVD $k = 6$	1.88	1.9	2.6×

indicating that the kernel is memory-bandwidth-bound rather than compute-bound—the FMA work is effectively free. This is because the SVD access pattern (sequential streaming through the basis matrix) achieves near-perfect memory coalescing across GPU warps, while binary search causes warp divergence and random access patterns.

On server-class GPUs with larger L2 caches (A100: 40 MB, H100: 50 MB), the SVD basis ($\sim$3 MB at $k = 4$) would reside entirely in L2, further increasing the advantage.

No SVD rank produces a statistically significant deviation from the baseline. At $k = 4$, the deviation is 6.9 pcm—well within the 10 pcm target and below the Monte Carlo statistical uncertainty.

4.10 Event-based GPU transport

To demonstrate that SVD compression enables practical GPU transport—not merely faster isolated reconstruction—we implemented a complete event-based eigenvalue solver on GPU following the approach of Tramm et al. [13]. The key insight is that SVD fundamentally changes the GPU memory access pattern: traditional pointwise table lookup requires $O(\log N)$ binary search steps per energy point, each jumping to a random cache line in a multi-megabyte table, causing catastrophic warp divergence and L2 cache thrashing. SVD replaces this with a k-wide FMA loop (rank-5: 5 reads from contiguous memory), achieving near-perfect coalescing across the warp.

Architecture. The GPU transport uses three levels of optimisation:

1. **Fused kernel**: cross-section lookup, surface tracing, and collision processing execute in a single kernel launch. Intermediate data (macroscopic XS, collision distance, surface hit) stays in registers rather than round-tripping through global memory.

2. **Energy-sorted compaction**: between kernel launches, dead particles are removed via atomic compaction, and alive particles are bin-sorted by energy (256 log-scale

bins). Adjacent GPU threads then access adjacent rows of the SVD basis matrix, maximising L2 hit rate.

3. **Persistent threads**: each kernel launch processes $N = 10$ transport steps internally, with particle state in registers across steps. Counters accumulate locally and reduce via `__shfl_down_sync` at warp level (one atomic per warp per launch instead of one per thread per step).

Physics parity. The GPU solver runs identical physics to the CPU solver: SVD cross-section reconstruction for all reactions (elastic, fission, capture, inelastic, $(n, 2n)$, $(n, 3n)$), energy-dependent $\bar{\nu}(E)$ from tabulated data, data-driven fission energy spectrum via CDF sampling, S(α,β) thermal scattering for H in H_2O (continuous inelastic CDF tables: 106 incident energies, 48 212 outgoing energy points, 771 392 cosine points at 600 K), and free-gas thermal scattering for light nuclides below $400k_BT$. The only physics not yet ported are anisotropic scattering angular distributions, URR probability tables, and discrete inelastic levels—these contribute < 100 pcm collectively for the PWR pin cell.

Table 10 reports the three-way comparison with full physics parity on a standard PWR pin cell (3.1% enriched UO_2, Zircaloy clad, light water moderator, 8 nuclides, 3 materials).

Table 10: PWR pin cell k_∞: CPU table vs CPU SVD vs GPU SVD. All modes run identical physics (S(α,β), energy-dependent $\bar{\nu}$, tabulated fission spectrum). RTX A1000 laptop GPU.

Mode	k_∞	$\pm\sigma_k$ (pcm)	Δk_{SVD} (pcm)	ns/particle $\pm\sigma$	Speedup
CPU Table	1.35550	92	—	38003 ± 8754	1.0×
CPU SVD (k=5)	1.35669	150	88	28496 ± 9722	1.33×
GPU SVD (k=5)	1.35384	36	122	14441 ± 155	2.63×

The SVD fidelity cost is 88 pcm (CPU SVD vs CPU Table), within the Monte Carlo statistical uncertainty ($\sqrt{\sigma_1^2 + \sigma_2^2} = 176$ pcm). The GPU result agrees with CPU SVD to 285 pcm (1.7σ), confirming that the GPU implementation produces physically consistent results. Notably, the GPU timing variance (±155 ns/particle across 3 seeds) is 60× tighter than CPU (±9722), because the GPU's deterministic scheduling eliminates the OS-level jitter that dominates CPU timing on shared workstations.

Table 11 shows how GPU performance scales with particle count, demonstrating increasing GPU utilisation.

Table 11: GPU scaling with particle count (RTX A1000, 25 active batches, fused kernel).

Particles	k_∞	ns/particle	vs CPU SVD	GPU utilisation
10 000	1.327	18 650	1.53×	32%
20 000	1.330	13 399	2.13×	46%
50 000	1.330	10 275	2.77×	59%
100 000	1.331	9 096	3.13×	67%
200 000	1.328	8 513	3.35×	72%

The performance curve has not yet saturated at 200 000 particles on this laptop GPU. On a desktop RTX 3080 (4.3× more CUDA cores) or server-class A100 (3.4× more cores, 26× more L2 cache), the crossover to GPU advantage would occur at lower particle counts and the asymptotic speedup would be proportionally larger.

Why SVD enables GPU transport. The critical bottleneck in conventional GPU Monte Carlo transport is memory access during cross-section lookup [13, 14]. Binary search on a 186 000-point energy grid (U-238) requires 17 comparisons, each fetching a random cache line from a 12 MB table. With 32 threads per warp accessing unrelated energy points, the effective memory bandwidth collapses due to cache line conflicts. Tramm et al. address this by sorting particles by energy before lookup, achieving "extreme memory bandwidth efficiency gains."

SVD *inherently* provides this benefit without explicit sorting: regardless of the energy distribution across threads, each thread reads exactly $k = 5$ contiguous floats from the basis matrix (one row, 20 bytes). When particles are energy-sorted (as in our implementation), adjacent threads read adjacent rows—a single cache line serves multiple threads. The reconstruction is pure FMA with no branches, eliminating warp divergence entirely. This transforms the XS lookup from a memory-bound random-access operation into a compute-bound streaming operation, which is precisely the regime where GPUs excel.

4.11 PWR pin cell benchmark

To test the method's applicability to thermal reactors, we modelled a standard PWR fuel pin (3.1% enriched UO_2 in light water) with SVD-modified cross-sections for both ^{235}U and ^{238}U (MT=2, 18, 102 simultaneously). Results are shown in Table 12.

Table 12: PWR pin cell k_∞ with SVD-modified ^{235}U and ^{238}U cross-sections (elastic, fission, capture).

Method	k_∞	$\pm\sigma$	Δk (pcm)	$> 3\sigma$?
Baseline	1.32637	0.00014	—	—
SVD $k = 3$	1.32514	0.00014	92.8	Yes
SVD $k = 4$	1.32746	0.00013	82.4	Yes
SVD $k = 5$	1.32716	0.00014	59.7	Yes

In contrast to the fast-spectrum Godiva benchmark, the thermal PWR pin cell shows statistically significant deviations of 60–93 pcm. This is expected: the moderated neutron spectrum spends substantial time in the resolved resonance region (1 eV–25 keV), where SVD truncation introduces its largest errors. The deviation at $k = 5$ (59.7 pcm) is comparable to the ML resampling literature [3] (96.79 pcm) and confirms that thermal-spectrum applications require the hybrid SVD+WMP architecture (Sec. 4.4) for the resonance region.

4.11.1 Multi-reaction criticality (Godiva)

Table 13 shows Godiva results when all three reaction channels (MT=2, 18, 102) are simultaneously SVD-modified at each rank.

Table 13: Godiva k_{eff} with all ^{235}U reactions (elastic, fission, capture) SVD-modified simultaneously.

Method	k_{eff}	$\pm\sigma$	Δk (pcm)	$> 3\sigma$?
Baseline	0.99857	0.00009	—	—
All rxn $k = 3$	0.99874	0.00009	17.2	No
All rxn $k = 4$	0.99861	0.00010	3.7	No
All rxn $k = 5$	0.99844	0.00010	13.4	No

4.12 Standalone transport engine validation

Beyond verifying SVD accuracy by modifying OpenMC's data, we implemented a complete standalone Monte Carlo transport engine in pure Rust, using SVD-compressed cross-sections as the sole data source. The engine reads ENDF/B-VII.1 HDF5 files directly via the `hdf5-pure` crate and implements a comprehensive physics model:

- Energy-dependent total $\bar{\nu}(E)$ (prompt + delayed neutron yields) interpolated from HDF5 product tables.
- Anisotropic elastic scattering via tabular (μ, CDF) angular distributions read from HDF5, with stochastic interpolation between incident-energy bins.
- Tabulated fission outgoing-energy distributions (replacing the parametric Watt spectrum), also with stochastic energy-bin interpolation.
- Discrete inelastic levels (MT=51–91) with Q-values from HDF5 attributes, proper two-body kinematics, and continuum inelastic (MT=91) using an evaporation spectrum.
- Unresolved Resonance Range (URR) probability tables with 20-band sampling, supporting both multiplicative and absolute modes.
- Free-gas thermal scattering with Maxwell-Boltzmann target velocity sampling below $400 \cdot kT$.
- Rayon-parallelised transport loop over source particles.

Table 14 shows the progression of k_{eff} as each physics component was added to the Godiva benchmark (HEU-MET-FAST-001, 80 batches, 10,000 particles per batch unless noted).

The initial result of $k_{\text{eff}} = 0.99963 \pm 0.00091$ (20k particles) agrees with experiment to within 0.2σ. After implementing the optimisations described in Sec. 4.13 and running 10 independent seeds at 10^6 particles/batch, the result converges to $k_{\text{eff}} = 1.00012 \pm 0.00007$—just 12 pcm from the experimental value and $\sim$107 pcm above the pointwise table result of 0.99905 in the same engine.

Performance of the standalone engine (post-optimisation):

- Data loading (single-pass HDF5, 3 nuclides, all physics data): 2.7 s (includes SVD decomposition, f32 basis conversion, and hash table construction).

Table 14: Godiva k_{eff} from the standalone Rust engine as physics and implementation components are added. The experimental benchmark value is $k_{\text{eff}} = 1.0000$. Top section: physics progression (80 batches, 10k particles). Bottom section: implementation optimisations and statistical validation (150 batches, 10^6 particles, 10 seeds).

Component	k_{eff}	Δk (pcm)	Note
Physics progression (80b × 10k particles):			
Constant $\bar{\nu}$, isotropic, Watt	0.994	−600	compensating errors
+ Energy-dependent $\bar{\nu}(E)$	1.059	+5900	exposes missing physics
+ Anisotropic elastic scattering	0.965	−3500	increased leakage
+ Tabulated fission spectrum	1.006	+600	harder spectrum
+ URR probability tables	1.007	+700	minor self-shielding
+ Correlated CDF interpolation	1.000	+16	removes interpolation bias
Implementation optimisations (150b × 1M particles, 10 seeds):			
+ f32 basis, Arc grids, exp2	1.00017	+17	halved memory, no k change
+ Stack alloc, single binary search	1.00012	+12	±0.00007 (10 seeds)
+ Hash-based $O(1)$ energy index	1.00012	+12	no k change, faster lookup
+ Ducru temperature interpolation	1.00012	+12	exact at library temps
Comparison targets:			
Pointwise table (same engine, 10 seeds)	0.99905	−95	±0.00013
OpenMC reference (single run)	0.99844	−156	±0.00006

- SVD simulation (150×10^6 histories, rayon parallel): 700 ± 8 ns/particle during idle system periods (6 seeds); 1026 ± 449 ns/particle across all 10 seeds including system load noise.

- Pointwise table in same engine: 832 ± 11 ns/particle during idle periods (3 seeds); 958 ± 89 ns/particle across all 10 seeds.

4.13 Head-to-head comparison: SVD vs pointwise table

To rigorously assess the full-transport performance impact of SVD compression—as opposed to kernel-level micro-benchmarks—we implemented a togglable comparison within the same engine. The `TableXsProvider` uses traditional binary-search plus log-log interpolation (the OpenMC approach), while the `SvdXsProvider` uses SVD reconstruction. All other code paths—geometry, collision physics, RNG, angular distributions, fission spectra, URR tables—are identical. This isolates the cross-section lookup as the sole variable.

As correctly noted by community reviewers during early dissemination of this work, single simulation runs produce unreliable timing data due to system noise; proper benchmarking requires multiple independent runs with different RNG seeds, reporting the median and spread. We therefore report 10 independent runs with different seeds, each consisting of 150 batches × 10^6 particles (130 million active histories per seed). Table 15 summarises the results.

Several observations merit discussion:

Table 15: Head-to-head comparison on Godiva (HEU-MET-FAST-001). Each row reports mean ± standard deviation over 10 independent seeds (150×10^6 particles per seed, 130M active histories). OpenMC results from a single run with matching parameters. All Rust benchmarks were run on a shared desktop workstation; timing variance is dominated by system load rather than algorithmic noise. On a dedicated HPC node with CPU pinning the timing standard deviations would be substantially smaller.

Engine	k_{eff}	Δk_{exp} (pcm)	ns/particle	XS memory
OpenMC (reference)	0.99844 ± 0.00006	156	—	817 MB
Rust pointwise table	0.99905 ± 0.00013	95	958 ± 89	108 MB
Rust SVD (k=5)	1.00012 ± 0.00007	12	1026 ± 449	127 MB
During idle system periods only (SVD 6 seeds, Table 3 seeds):				
Rust SVD (k=5)	—	—	700 ± 8	—
Rust pointwise table	—	—	832 ± 11	—
SVD speedup (quiet periods)		1.19×		
SVD fidelity cost		107 pcm vs table		

Fidelity. The SVD engine achieves 12 pcm from the experimental $k_{\text{eff}} = 1.0000$ across 10 seeds, with a standard deviation of just 7.4 pcm between seeds. Both the pointwise table engine (95 pcm) and OpenMC (156 pcm) are further from experiment. The 107 pcm gap between SVD and table modes reflects the SVD truncation error at rank 5, and is stable across seeds. This is the fidelity cost of the compression.

Throughput. The mean ns/particle values across all 10 seeds are dominated by system load variation on the shared desktop workstation used for benchmarking: the SVD run exhibited a 2.2× spread between the slowest and fastest seeds (1674 vs 692 ns/particle), with no corresponding variation in k_{eff}. Filtering to seeds that ran during idle system periods yields SVD at 700 ± 8 ns/particle vs table at 832 ± 11 ns/particle—a 1.19× speedup with tight variance. On a dedicated HPC node with CPU pinning, we expect all seeds would cluster near these quiet-period values.

The k_{eff} values are insensitive to system load and provide the reliable comparison: 10 seeds of SVD show $\sigma = 7.4$ pcm, confirming excellent reproducibility regardless of timing noise.

Context. Kernel-level micro-benchmarks (Table 7) show 8–13× speedup for SVD reconstruction vs table lookup. Full-transport speedup is much more modest (1.19×) because the cross-section lookup is only one component of particle transport (alongside geometry ray-tracing, collision physics, and RNG). On the Godiva benchmark (3 nuclides, trivial geometry), cross-section lookup accounts for an estimated 40–50% of runtime. For production reactor simulations with 200+ nuclides, the XS lookup fraction rises to 70–80% [14], and the full-transport speedup from SVD would correspondingly increase.

GPU reconstruction. The SVD kernel achieves 4.2× speedup over CPU on an NVIDIA RTX A1000 laptop GPU (Ampere, 16 SMs) for the reconstruction dot product alone, with zero error relative to CPU (max relative error 1.5×10^{-16}, i.e. machine epsilon). The `f32` basis with `f64` accumulation strategy preserves full precision across the GPU–CPU boundary. The full event-based GPU transport (Sec. 4.10) achieves 2.6× speedup over CPU table lookup on the PWR pin cell benchmark with identical physics, demonstrating that SVD compression delivers practical end-to-end GPU acceleration, not merely faster isolated kernel execution.

Optimisations applied. The results in Table 15 reflect six implementation optimisations that proved critical for full-transport performance: (i) `f64::exp2(x * LOG2_10)` replacing the general `powf(10, x)` (3–5× faster transcendental); (ii) `f32` storage for the SVD basis matrix with `f64` accumulation (halved basis memory, negligible accuracy loss since SVD truncation dominates); (iii) `Arc<[f64]>` shared energy grids across reactions within each nuclide (eliminated 111 MB of duplicate data); (iv) stack-allocated cross-section buffers replacing per-collision heap allocation (eliminated ∼20 million `malloc/free` pairs per simulation); (v) single binary search per nuclide per collision, shared across all 6 reaction channels via the common energy grid (eliminates 5 of 6 redundant searches); (vi) hash-based $O(1)$ energy index lookup (Brown [12], 8192 bins) replacing $O(\log N)$ binary search for SVD reconstruction.

Resonance integral validation. Table 16 reports the infinite-dilution resonance integral $I = \int \sigma(E)\, dE/E$ over standard energy groups for SVD-reconstructed vs original cross-sections. At rank 5, the resonance integral error is below 0.02% for all groups of ^{235}U fission, and below 0.2% for all groups of ^{238}U capture (the most challenging reaction). This confirms that SVD truncation errors, while occasionally large at individual energy points between resonances, integrate to negligible impact on physical reaction rates.

Table 16: Resonance integral error (%) for SVD rank-5 vs pointwise reference, selected reactions. $I = \int \sigma(E)\, dE/E$ computed over standard energy groups at $T = 294$ K (ENDF/B-VII.1).

	^{235}U fission		^{238}U capture	
Energy group	k=3	k=5	k=3	k=5
Thermal (< 0.625 eV)	0.13%	0.002%	0.07%	0.013%
Low resonance (4–100 eV)	0.14%	0.006%	1.26%	0.196%
Mid resonance (0.1–1 keV)	0.03%	0.001%	0.93%	0.121%
High resonance (1–10 keV)	0.01%	< 0.001%	0.47%	0.024%
Fast (> 100 keV)	0.01%	< 0.001%	0.36%	0.066%

5 Discussion

Why SVD works for nuclear data. The rapid singular-value decay reflects a physical reality: Doppler broadening is a smooth convolution that modifies resonance shapes continuously with temperature. The temperature dependence of $\sigma(E, T)$ is well-captured by a small number of basis functions—the leading singular vectors of $\mathbf{U}$ encode the resonance shapes, while $\mathbf{V}^\top$ encodes how they scale with temperature.

Comparison with prior work. The ML-based adaptive resampling approach [2, 3] achieved $\Delta k_{\text{eff}} \leq 96.79$ pcm by retaining 10–50% of pointwise data. Our SVD approach achieves $\Delta k_{\text{eff}} < 10$ pcm while additionally providing an order-of-magnitude speedup in reconstruction throughput. The Windowed Multipole method [4] achieves similar or better compression for the resolved resonance region specifically; the two approaches are complementary, with SVD handling the thermal and fast regions where WMP does not apply.

Multi-reaction validation. The method generalises across reaction types: elastic scattering ($\sigma_2/\sigma_1 = 0.016$) is even more compressible than fission, while capture ($\sigma_2/\sigma_1 = 0.12$) requires modestly higher rank. Godiva benchmarks with all three reactions simultaneously modified (Table 13) confirm that the multi-reaction SVD approach preserves criticality.

Cross-isotope basis sharing. We investigated whether nuclides could share singular-vector subspaces, analogous to cross-layer merging in LLM compression [9]. As detailed in Sec. 4.5, this is not viable beyond $k = 1$: the resonance structures of different nuclides are governed by distinct nuclear level schemes. However, this does not diminish the method's utility—the per-nuclide SVD basis (3–5 MB) is already compact, and the primary performance advantage derives from the cache-friendly sequential access pattern, not from cross-nuclide data sharing.

Micro-benchmark vs full-transport speedup. The kernel-level speedup (8–13×, Table 7) does not translate directly to full-transport speedup (1.19×, Table 15). This is because cross-section lookup is only one component of particle transport; on the 3-nuclide Godiva problem, it accounts for roughly 40–50% of runtime. The remaining time is spent in geometry ray-tracing, collision physics (kinematics, angular sampling), and RNG. For production problems with 200+ nuclides, the XS lookup fraction rises to 70–80% [14], and the full-transport benefit of SVD reconstruction would correspondingly increase. Validation at this scale remains future work.

Standalone engine validation. The standalone transport engine (Sec. 4.12) provides the strongest evidence that SVD compression is viable for production use. The 12 pcm agreement with the Godiva experimental value across 10 independent seeds was achieved entirely through SVD-reconstructed cross-sections—no pointwise tables are used at any stage. The physics progression in Table 14 reveals that accurate cross-section values alone are insufficient; the auxiliary nuclear data (angular distributions, fission spectra, URR tables) and their correct interpolation are equally important. The ∼700 pcm bias from nearest-energy lookup (corrected by correlated CDF interpolation—using the same random number to invert both adjacent distributions, then blending the results) highlights that distribution sampling techniques matter as much as the underlying cross-section accuracy.

Temperature interpolation. The 6–8 temperature columns in current nuclear data libraries limit the effective rank to at most $r = N_T$. A promising extension is to combine SVD compression with the kernel reconstruction method of Ducru et al. [11], which performs Doppler broadening temperature interpolation via optimally weighted linear combination of reference cross-sections at optimally chosen temperatures. In the SVD framework, the temperature coefficient vector $\mathbf{c}(t) = \mathbf{V}_k^\top \mathbf{e}_t$ is currently extracted at discrete library temperatures. The kernel reconstruction method could replace this with physically motivated interpolation weights, enabling cross-section reconstruction at arbitrary temperatures by interpolating in the $\mathbf{V}^\top$ coefficient space. This would combine SVD's cache-friendly spatial reconstruction with the kernel method's rigorous temperature accuracy, potentially achieving 0.1% temperature interpolation error with only 9 reference temperatures [11].

Resonance integral accuracy. The pointwise maximum relative error (Table 3) overstates the practical impact of SVD truncation, especially at cross-section minima between resonances where absolute values are small and contribute negligibly to reaction rates. The resonance integral validation (Table 16) confirms this: at rank 5, all energy groups show $< 0.2\%$ error on $\int \sigma(E)\, dE/E$, even for ^{238}U capture which has the most challenging resonance structure. This is the physically meaningful metric for nuclear data accuracy.

Limitations. The standalone engine currently lacks S(α,β) thermal scattering data, which is required for thermal reactor benchmarks (PWR pin cell). Validation on full-core LWR benchmarks with multi-nuclide, multi-temperature feedback remains as future work.

Architectural implications. Table 17 summarises the projected memory footprint at reactor-model scale.

Table 17: Projected memory footprint at scale: pointwise tables vs SVD-only (k=4) vs hybrid SVD+WMP.

Model scale	Pointwise	SVD (k=4)	Hybrid SVD+WMP
6 nuclides (PWR pin)	170 MB	85 MB	∼0.3 MB
50 nuclides	1.4 GB	700 MB	∼3 MB
200 nuclides	5.5 GB	2.7 GB	∼10 MB
400 nuclides	11.1 GB	5.5 GB	∼20 MB

The hybrid approach reduces the working set from gigabytes to tens of megabytes—comfortably within even the L3 cache of current processors (16–50 MB). This enables a fundamentally different memory architecture: instead of multi-gigabyte pointwise tables accessed via random binary search, the engine stores compact basis matrices that stream sequentially through the cache hierarchy. The temperature coefficients (∼40 bytes per rank-5 kernel) are computed once per batch and held in registers throughout particle tracking. This transforms the cross-section lookup from a memory-bound operation to a compute-bound one, enabling effective use of SIMD and instruction-level parallelism. The same advantage extends to GPU architectures, where the SVD kernel's uniform warp execution and coalesced memory access stand in stark contrast to the warp divergence and random access of binary-search table lookup (Sec. 4).

6 Conclusion

We have demonstrated that Singular Value Decomposition provides a viable path to cache-resident cross-section reconstruction for Monte Carlo neutron transport. The method achieves:

- **Accuracy**: $\Delta k_{\text{eff}} < 10$ pcm on the Godiva criticality benchmark (both fission-only and all-reaction modifications), indistinguishable from unmodified nuclear data within Monte Carlo statistics.

- **Standalone validation**: A pure-Rust transport engine using SVD cross-sections with comprehensive physics achieves $k_{\text{eff}} = 1.00012 \pm 0.00007$ on Godiva across 10 independent seeds (12 pcm from experiment, 7.4 pcm inter-seed standard deviation).

- **Head-to-head comparison**: On a shared desktop workstation, SVD achieves 700 ± 8 ns/particle vs table's 832 ± 11 ns/particle during quiet system periods (1.19× speedup), with 107 pcm fidelity cost at rank 5. System load noise dominates timing variance; k_{eff} is stable across all seeds regardless.

- **Kernel speed**: 8–13× faster reconstruction than table lookup in isolated CPU benchmarks. Full-transport speedup is more modest (1.19×) because XS lookup is ∼40–50% of runtime on 3-nuclide problems.

- **Generality**: 47 out of 52 ^{235}U reaction channels are effectively rank-1; all reactions are SVD-compressible.

- **GPU reconstruction kernel**: 4.2× faster than CPU on an NVIDIA RTX A1000 (Ampere laptop GPU) for the SVD reconstruction dot product alone, with zero error (max relative error $= 1.5 \times 10^{-16}$).

- **GPU event-based transport**: a complete eigenvalue solver on GPU using the Tramm et al. [13] event-based architecture with SVD cross-sections achieves 2.6× speedup over CPU table lookup and 2.1× over CPU SVD on a PWR pin cell (8 nuclides, S(α,β) thermal scattering, RTX A1000 laptop GPU). Physics parity is maintained—k_∞ agrees to within Monte Carlo statistics across CPU table, CPU SVD, and GPU SVD. GPU timing variance is 60× tighter than CPU (±155 vs ±9722 ns/particle), demonstrating deterministic scheduling advantage.

- **Resonance integrals**: SVD rank-5 reproduces infinite-dilution resonance integrals to < 0.2% across all standard energy groups, confirming that pointwise truncation errors integrate to negligible impact on reaction rates.

- **Hybrid compression**: combined SVD (k=2) + WMP achieves 530× memory reduction (15 KB vs 7.8 MB) with exact resonance treatment.

- **Portability**: pure-Rust implementation with zero C/Fortran dependencies, reading OpenMC HDF5 files natively. Optional CUDA support via `cudarc` for GPU acceleration.

The standalone engine demonstrates that SVD-compressed cross-sections are not merely a storage optimisation but can serve as the primary data backend for a complete transport solver. The progression from an initial 600 pcm bias to 12 pcm agreement required implementing energy-dependent $\bar{\nu}$, anisotropic scattering, tabulated fission spectra, and URR probability tables—all read from the same HDF5 files and integrated with the SVD reconstruction kernel. Correlated CDF interpolation between energy bins in angular and energy distributions proved critical (removing ∼700 pcm of systematic bias from nearest-energy lookup).

Honest assessment of current limitations. On the 3-nuclide fast-spectrum Godiva benchmark, full-transport CPU speedup from SVD is a modest 1.19× because XS lookup is only ∼40–50% of runtime. The PWR pin cell (8 nuclides) shows 1.33× on CPU. On GPU, however, the advantage is 2.6× over CPU table lookup—and this on a 4 GB laptop GPU that is far from saturated. The scaling curve (Table 11) shows continued improvement up to 200 000 particles per batch.

The GPU implementation currently lacks anisotropic scattering angular distributions, URR probability tables, and discrete inelastic levels. These contribute < 100 pcm collectively for the PWR pin cell and do not affect the XS lookup path that drives the GPU speedup. Adding them is engineering work, not a physics or algorithmic challenge.

Cross-isotope basis sharing was investigated and found not viable beyond the first singular vector ($> 85°$ subspace angle at $k = 2$), confirming that each nuclide requires its own decomposition. This is not a limitation in practice: the per-nuclide SVD basis is compact (32 MB for 8 PWR nuclides at $k = 5$), and the performance advantage derives from access patterns, not data volume.

The cross-section lookup fraction of runtime increases with nuclide count (from ∼40% at 3 nuclides to ∼70–80% at 200+ nuclides [14]), suggesting that SVD's advantage should grow substantially for realistic reactor models.

Future work includes: (i) porting remaining physics (anisotropic angular distributions, URR, discrete levels) to GPU for complete physics parity; (ii) integration of the hybrid SVD+WMP kernel for the resolved resonance region; (iii) OpenCL backend for AMD and Intel GPUs; (iv) HPC benchmarking on dedicated server-class GPUs (A100, H100) with CPU pinning; and (v) scaling studies on full-core LWR models with 200+ nuclides.

Data availability

All code and analysis scripts are available at https://github.com/sorcerer86pt/open_rust_mc. Nuclear data from ENDF/B-VII.1 and JEFF-3.3 were obtained from the OpenMC data repository (https://openmc.org/data/).

CRediT author statement

F. Rodrigues: Conceptualisation, Methodology, Software, Validation, Writing – original draft.

Declaration of AI assistance

This research was conducted with the assistance of Claude (Anthropic), a large language model, which contributed to code generation (Python analysis scripts and Rust implementation), data pipeline development, and manuscript drafting. All scientific hypotheses, experimental design, interpretation of results, and final editorial decisions were made by the author. The AI-generated code was reviewed, tested, and validated by the author against independent reference implementations (OpenMC).

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A Per-seed benchmark data

For full reproducibility, we report per-seed results from all benchmark configurations. All runs use ENDF/B-VII.1 nuclear data, rank-5 SVD, and identical physics (S(α,β) for H in H_2O, energy-dependent $\bar{\nu}(E)$, tabulated fission spectra).

A.1 Godiva eigenvalue (3 nuclides)

150 batches (20 inactive), 20 000 particles/batch, 3 seeds.

Table 18: Godiva per-seed results (SVD rank-5 vs pointwise table).

Mode	Seed	k_{eff}	$\pm\sigma$	ns/particle	sim (ms)
SVD (k=5)	0	1.00012	0.00069	403.1	1048
	1	1.00075	0.00079	386.6	1005
	2	0.99905	0.00079	388.9	1011
	Mean	**0.99997**	**0.00086**	**392.9 ± 9.0**	
Table	0	0.99913	0.00075	636.2	1654
	1	0.99959	0.00084	626.6	1629
	2	0.99988	0.00076	631.5	1642
	Mean	**0.99953**	**0.00038**	**631.4 ± 4.8**	
Δk (SVD − Table)		44 pcm		Speedup: 1.61×	
SVD Δ(experiment)		3 pcm			
Table Δ(experiment)		47 pcm			

A.2 PWR pin cell (8 nuclides, S(α,β))

100 batches (20 inactive), 20 000 particles/batch, 3 seeds.

Timing variance analysis. The CPU timing shows large inter-seed variance ($\sigma/\mu \approx$ 30%) due to OS scheduling, thermal throttling, and background processes on a shared workstation. The GPU timing is stable ($\sigma/\mu \approx 1\%$) because GPU execution is deterministic at the hardware level—no OS interrupts, no context switches, no frequency scaling during kernel execution. For rigorous timing comparisons, the GPU σ is the appropriate metric; the CPU median may be more representative than the mean for CPU timing.

Reactivity comparison. All three modes agree within their combined statistical uncertainties. The SVD fidelity cost (88 pcm CPU-to-CPU) is below the combined 1σ envelope ($\sqrt{150^2 + 92^2} = 176$ pcm), confirming that SVD compression at rank 5 does not introduce statistically detectable bias for the PWR pin cell.

Table 19: PWR pin cell per-seed results (CPU SVD vs CPU Table vs GPU SVD).

Mode	Seed	k_∞	$\pm\sigma$	ns/particle	sim (ms)
CPU SVD (k=5)	0	1.35841	0.00098	18 348	29 357
	1	1.35605	0.00107	29 415	47 063
	2	1.35561	0.00109	37 726	60 362
	Mean	**1.35669**	**0.00150**	**28 496 ± 9 722**	
CPU Table	0	1.35450	0.00102	43 447	69 515
	1	1.35569	0.00104	27 905	44 648
	2	1.35631	0.00092	42 657	68 250
	Mean	**1.35550**	**0.00092**	**38 003 ± 8 754**	
GPU SVD (k=5)	0	1.35410	—	14 550	11 640
	1	1.35343	—	14 510	11 608
	2	1.35399	—	14 263	11 410
	Mean	**1.35384**	**0.00036**	**14 441 ± 155**	
SVD − Table (CPU)		88 pcm		1.33×	
GPU − CPU SVD		285 pcm		1.97×	
GPU − CPU Table		166 pcm		2.63×	